

EDUCATION

Wuhan University | State Key Lab. LIESMARS

Wuhan, China | 2023.09 - 2026.06

M.Sc. in Cartography and Geographic Information Systems

- GPA: 3.55/4.00, Average Score: 89.40/100, IELTS: 7.0(6.0)
- Research area: Computer Vision and 3D Reconstruction

Wuhan University | School of Remote Sensing and Information Engineering

Wuhan, China | 2019.09 - 2023.06

B.Eng. in Remote Sensing Science and Technology

- GPA: 3.75/4.00, Average Score: 88.90/100, Recommended Exemption

PUBLICATIONS

1. **Xu Pan**, Zhenglin Wan, Xingrui Yu, Xianwei Zheng, Youkai Ke, Ming Sun, Rui Wang, Ziwei Wang, Ivor Tsang. **SA-VLA: Spatially-Aware Reinforcement Learning for Flow-Matching Vision-Language-Action Models**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2026.
2. **Xu Pan**, Zimin Xia, Xianwei Zheng. **Scale-aware Co-visible Region Detection for Image Matching**. *ISPRS Journal of Photogrammetry and Remote Sensing [JCR Q1, IF 12.2]*, 2025. <https://doi.org/10.1016/j.isprsjprs.2025.08.015>
3. **Xu Pan**, Zhenglin Wan, Xingrui Yu, Xianwei Zheng, Youkai Ke, Ming Sun, Rui Wang, Ziwei Wang, Ivor Tsang. **SFM: Spatially-Aware Flow Matching for Embodied Reinforcement Learning**. *Under Review*, 2026. Available at <https://arxiv.org/abs/2602.00743>.
4. Ming Sun, Rui Wang, Xingrui Yu, Lihua Jing, Hangyu Du, Zhenglin Wan, **Xu Pan**, Ivor Tsang. **Towards Backdoor-Based Ownership Verification for Vision-Language-Action Models**. *Under Review*, 2026. Available at <https://arxiv.org/abs/2605.09005>
5. **Xu Pan**, Qiyuan Ma, Mingyue Dong, He Chen, Wei Ji, Xianwei Zheng. **SAMatcher: Co-Visibility Modeling with Segment Anything for Robust Feature Matching**. *IEEE Transactions on Image Processing (Under Review)*, 2026. Available at <https://arxiv.org/abs/2606.03406>
6. Jiashen Huang, Yu Jia, **Xu Pan**. **Trusting AI in China: Institutional Trust and the Domestic Advantage of DeepSeek over ChatGPT**. *Chinese Journal of Communication (Under Review)*, 2026. Available at <https://arxiv.org/abs/2606.01228>.

EXPERIENCES

Nanyang Technological University (NTU)

Singapore | 2026.07 - present

Research Assistant | *School of Electrical and Electronic Engineering*

- Under the supervision of Prof. Ziwei Wang, at Perception and Embodied Intelligence (PINE) Lab

Agency for Science, Technology and Research (A*STAR)

Singapore | 2025.08 - 2026.06

Research Assistant | *Centre for Frontier AI Research (CFAR), Institute of High Performance Computing (IHPC)*

- Pioneered spatially-aware reinforcement learning adaptation for flow-based VLA models, preserving implicit spatial inductive bias through fused geometric prior and preventing degradation under distribution shifts.
- Designed geometry-aligned dense rewards and spatially-conditioned exploration, enabling stable RL fine-tuning, superior zero-shot spatial generalization, and robust transferable behaviors in complex manipulation scenes.

Baidu International Technology (Shenzhen) Co., Ltd.

Shenzhen, China | 2024.12 - 2025.03

Research Intern | *Foundation Public Technology Group, International Technology R&D Department*

- Developed DiT-based framework for controllable, arbitrarily long video generation via key-frame conditioning.
- Improved the visual fidelity of rendered text in image generation by designing specialized attention mechanisms.
- Contributed to the **SynClub** AI chat app through assessment of frontier multimodal generative models and optimization of text-to-image/video modules, delivering improved performance and higher-quality outputs.

PROJECTS

Intelligent Interpretation Platform for Remote Sensing Images Based on Baidu PaddlePaddle	2022.03 - 2022.08
 TwSphinx54/I2RSI	
Web-based Global Natural Disaster Assessment System	2022.05 - 2022.09
 TwSphinx54/GNDAS	

ACADEMIC ACTIVITIES

2026 Vision and Learning Seminar (Valse), China Society of Image and Graphics (CSIG) Wuhan, China	2026.05.08 - 05.10
2025 Annual Conference on Photogrammetry and Remote Sensing, Chinese Society for Geodesy, Photogrammetry and Cartography (CSGPC) Kunming, China	2025.07.28 - 07.30

AWARDS AND HONORS

• Outstanding Graduating Graduate Student , Wuhan University	2026.04
• Outstanding Graduate Student , Wuhan University	2025.11
• Graduate Academic Excellence Scholarship , Wuhan University	2025.11
• Graduate Academic Excellence Scholarship , Wuhan University	2024.11
• Outstanding Student Leader , Wuhan University	2024.10
• Outstanding Student Club Leader , Wuhan University	2024.08
• Active Contributor to Social Activities , School of Remote Sensing and Information Engineering	2023.05
• Wuhan University Class C Scholarship , Wuhan University	2022.10
• National Second Prize , China Software Cup College Student Software Design Competition	2022.08
• Honorable Mention , Mathematical Contest In Modeling	2022.03
• Third Prize , Asia and Pacific Mathematical Contest in Modeling	2021.12
• First Prize in Hubei Division , China Undergraduate Mathematical Contest in Modeling	2021.10
• Outstanding Student , Wuhan University	2021.09
• Bronze Medal , China Collegiate Algorithm Design & Programming Challenge Contest	2021.03
• Second Prize in Final , Translation & Interpreting Contest of Hubei Province	2020.05

SKILLS

Languages: Mandarin, English (IELTS 7.0)

Programming: Python, C++, MATLAB, HTML, CSS, JavaScript, C#, SQL, R

Soft Skills: Leadership, Project Management, Event Organization, Public Speaking, Design & Editing

CERTIFICATES

System Integration Project Management Engineer

Qualification Certificate of Computer and Software Technology Proficiency, MOHRSS & MIIT

Certificate of Level 4 in Network Engineer

National Computer Rank Examination, NEEA

Certificate of Level 3 in Network Technology

National Computer Rank Examination, NEEA

Certificate of Level 3 in Database Technology

National Computer Rank Examination, NEEA

ACADEMIC SERVICES

Reviewer 10th Annual Conference on Robot Learning (CoRL'26)
Member ISPRS Student Consortium (ISPRS SC)
Volunteer 2022/2023 International Graduate Workshop on Geo-Informatics
The 21st China Ocean Color Conference

LEADERSHIP

Chairman The 22nd LIESMARS Graduate Student Union
Director GeoScience Café Operation Center